

PGHN-S-26-00112.pdf

By Sultan Toruntju

Pediatric Gastroenterology, Hepatology & Nutrition

Sea Hare-Papaya Syrup and Iron Status in Stunted Toddlers

--Manuscript Draft--

Manuscript Number:	
Full Title:	Sea Hare-Papaya Syrup and Iron Status in Stunted Toddlers
Article Type:	Original Article
Section/Category:	Nutritional diseases
Keywords:	Child Nutrition Disorders; Anemia; Hemoglobins; Ferritins; Growth Disorders; Carica; Child, Preschool
Manuscript Region of Origin:	INDONESIA
Abstract:	Purpose: Stunting may coexist with anemia and impaired iron status in early childhood. This study examined hemoglobin and serum ferritin outcomes after a food-based syrup containing sea hare (<i>Dolabella auricularia</i>) extract and papaya in stunted toddlers. Materials and Methods: This secondary analysis used data from a five-arm pretest-posttest controlled community nutrition intervention conducted in Southeast Sulawesi, Indonesia, from September 2023 to July 2024. Fifty stunted toddlers were allocated to three sea hare-papaya syrup groups (80:20, 70:30, or 60:40), a positive-control group receiving commercial iron syrup, or a placebo group. The intervention was given every two days for 2 weeks. Hemoglobin and serum ferritin were measured before and after intervention. Dietary intake was assessed using a semi-quantitative food-frequency questionnaire. Results: Among 50 enrolled toddlers, 38 (76.0%) had anemia at baseline. Hemoglobin increased significantly in the three sea hare-papaya groups: 10.9 ± 0.61 to 11.7 ± 0.51 g/dL ($P=0.002$), 11.1 ± 0.86 to 11.8 ± 0.57 g/dL ($P=0.008$), and 10.8 ± 1.23 to 11.6 ± 0.45 g/dL ($P=0.015$). Changes in the control groups were not significant. Serum ferritin increased numerically but did not change significantly in any group. Conclusion: Sea hare-papaya syrup was associated with within-group hemoglobin improvement in stunted toddlers, but ferritin responses were inconclusive. Larger registered trials with standardized dosing and stronger between-group analyses are needed.
Short Title:	Sea Hare-Papaya Syrup in Toddlers

Sea Hare-Papaya Syrup and Iron Status in Stunted Toddlers

ABSTRACT

Purpose: Stunting may coexist with anemia and impaired iron status in early childhood. This study examined hemoglobin and serum ferritin outcomes after a food-based syrup containing sea hare (*Dolabella auricularia*) extract and papaya in stunted toddlers. **Materials and Methods:** This secondary analysis used data from a five-arm pretest-posttest controlled community nutrition intervention conducted in Southeast Sulawesi, Indonesia, from September 2023 to July 2024. Fifty stunted toddlers were allocated to three sea hare-papaya syrup groups (80:20, 70:30, or 60:40), a positive-control group receiving commercial iron syrup, or a placebo group. The intervention was given every two days for 6 weeks. Hemoglobin and serum ferritin were measured before and after intervention. ² Dietary intake was assessed using a semi-quantitative food-frequency questionnaire.

Results: Among 50 enrolled toddlers, 38 (76.0%) had anemia at baseline. Hemoglobin increased significantly in the three sea hare-papaya groups: 10.9 ± 0.61 to 11.7 ± 0.51 g/dL ($P=0.002$), 11.1 ± 0.86 to 11.8 ± 0.57 g/dL ($P=0.008$), and 10.8 ± 1.23 to 11.6 ± 0.45 g/dL ($P=0.015$). Changes in the control groups were not significant. Serum ferritin increased numerically but did not change significantly in any group. **Conclusion:** Sea hare-papaya syrup was associated with within-group hemoglobin improvement in stunted toddlers, but ferritin responses were inconclusive. Larger registered trials with standardized dosing and stronger between-group analyses are needed.

Keywords: Child Nutrition Disorders; Anemia; Hemoglobins; Ferritins; Growth Disorders; Carica; Child, Preschool

INTRODUCTION

21

Stunting in early childhood reflects chronic undernutrition and remains a major public health problem in low- and middle-income countries. Beyond impaired linear growth, stunting is associated with poor dietary quality, recurrent infection, delayed development, and increased vulnerability to micronutrient deficiency.¹ These problems are particularly important during the first years of life, when nutrient inadequacy may affect hematopoiesis, immune function, and neurodevelopment.

Anemia is common among young children and may coexist with stunting. Hemoglobin concentration provides a practical indicator of oxygen-carrying capacity, while serum ferritin is widely used to assess iron stores.²⁻⁵ However, ferritin interpretation can be complicated by infection or inflammation, and changes in ferritin may require a longer intervention period than changes in hemoglobin.^{2,5,6}

Food-based interventions using locally available resources may provide complementary approaches to improve child nutrition. Sea hare (*Dolabella auricularia*), locally referred to as sea rabbit, has been reported to contain secondary metabolites, including flavonoids, alkaloids, terpenoids, saponins, and tannins.⁷ Papaya may contribute nutrients and bioactive components relevant to dietary quality and hematopoiesis.⁸⁻¹⁰ Evidence on the combined use of sea hare extract and papaya for hematological outcomes in stunted toddlers remains limited.

The present manuscript reports hemoglobin and serum ferritin outcomes from a five-arm community nutrition intervention in stunted toddlers. Growth and motor-development outcomes from the same source cohort have been reported previously; the hematological outcomes presented here were not the focus of that earlier publication.¹¹ This secondary analysis aimed to examine changes in hemoglobin and serum ferritin after sea hare-papaya syrup administration and to interpret the findings cautiously in light of sample size and trial-reporting limitations.

MATERIALS AND METHODS

Study design and setting

This secondary analysis used data from a five-arm pretest-posttest controlled community nutrition intervention conducted at integrated community health posts in a primary health care setting in Southeast Sulawesi, Indonesia, from September 2023 to July 2024. The source study evaluated honey-based syrup formulations containing sea hare extract and papaya in stunted and malnourished toddlers. The available study record described allocation into five intervention arms and masking of syrup composition, but complete documentation of random sequence generation and allocation concealment was not available. Therefore, the present report is framed as an exploratory controlled intervention analysis rather than as evidence of intervention superiority.

Participants

Toddlers identified as stunted and malnourished through five community health posts were considered for enrollment. Children were eligible if they were clinically stable, had no acute respiratory infection or diarrhea at recruitment, had available baseline and follow-up laboratory data, and had ²² written informed consent from a parent or legal guardian. Children with chronic illness, severe medical conditions, or incomplete participation during the intervention period were excluded.

A total of 96 toddlers were screened for hemoglobin status. Fifty toddlers were included in the intervention phase and allocated equally to five groups of 10 participants. ²⁴ Anemia was defined as hemoglobin <11.0 g/dL and was analyzed as a baseline characteristic rather than as a mandatory enrollment criterion.

Interventions and masking

The three experimental groups received sea hare-papaya syrup formulations containing sea hare extract and papaya extract in ratios of 80:20, 70:30, or 60:40. The positive-control group received

commercial iron syrup, and the negative-control group received placebo. All formulations were prepared as honey-based syrups and administered every two days for 6 weeks.

Sea hare samples were collected from coastal areas of Southeast Sulawesi, Indonesia. Samples were repeatedly washed to reduce salt content and dried at temperatures below 50°C without direct sunlight exposure before extraction. Ripe papaya was purchased from a local market in Kendari, Indonesia, cleaned, cut, and dried using an electric oven at 40-50°C before extraction. Extract preparation was conducted in a laboratory in Kendari. The source records did not contain complete batch-level quantitative composition or active-compound standardization; this limitation was considered when interpreting the findings.

Laboratory personnel managed syrup preparation and group labeling. Investigators involved in field data collection were not informed of the exact formulation assigned to each group until completion of the intervention period. The available records did not document a formal adverse-event grading system. No serious adverse event was identified in the available study forms.

Outcome measures and covariates

The outcomes for this analysis were hemoglobin concentration and serum ferritin level before and after 6 weeks of intervention. Hemoglobin was expressed in g/dL and serum ferritin in ng/mL. Laboratory measurements were performed before and after intervention using routine clinical laboratory procedures. Dietary intake of energy, protein, and iron ² was assessed using a semi-quantitative food-frequency questionnaire, a digital dietary scale, NutriSurvey software, the Indonesian Food Composition Table, and the 2019 Recommended Dietary Allowance.

² Statistical analysis

Data were analyzed using SPSS version 20. Descriptive statistics were used to summarize baseline screening ²³ results and dietary intake. Paired t-tests were used to compare pre- and post-intervention hemoglobin and serum ferritin within each group. The original analysis included exploratory

between-group comparisons after intervention, but the source record did not provide sufficient detail to support a robust claim of superiority over either control group. Statistical significance was set at $P < 0.05$. No adjustment for multiple testing was applied; therefore, all inferential findings are interpreted as exploratory.

Ethical considerations

The study was approved by a Health Research Ethics Committee (Ethical Clearance No. 142/KEPK/IAKMI/XI/2020). Written informed consent was obtained from the parents or legal guardians of all participants. The trial was not prospectively registered.

RESULTS

Among 96 toddlers screened for hemoglobin, 38 (39.5%) had anemia and 58 (60.4%) had normal hemoglobin levels. Fifty toddlers entered the intervention phase, with 10 participants in each of the five study groups. At baseline, 38 of the 50 enrolled toddlers (76.0%) were anemic.

Baseline dietary intake was broadly comparable across groups (Table 1). Energy intake ranged from 1003 kcal in the 80:20 sea hare-papaya group to 1119 kcal in the placebo group. Protein intake ranged from 33.2 g in the commercial iron-syrup group to 41.3 g in the placebo group, while iron intake ranged from 8.6 mg in the placebo group to 10.6 mg in the 70:30 sea hare-papaya group.

Hemoglobin changes are shown in Table 2. Hemoglobin increased significantly in all three sea hare-papaya syrup groups after 6 weeks. Mean hemoglobin increased from 10.9 ± 0.61 to 11.7 ± 0.51 g/dL in the 80:20 group ($P = 0.002$), from 11.1 ± 0.86 to 11.8 ± 0.57 g/dL in the 70:30 group ($P = 0.008$), and from 10.8 ± 1.23 to 11.6 ± 0.45 g/dL in the 60:40 group ($P = 0.015$). The changes in the commercial iron-syrup group and placebo group were not statistically significant.

Serum ferritin changes are shown in Table 3. Ferritin increased numerically in each group, including the three sea hare-papaya groups and both control groups, but none of the within-group changes reached statistical significance. The available post-intervention between-group

comparisons did not show statistically significant differences for hemoglobin or ferritin; therefore, intervention superiority over control treatment was not inferred.

DISCUSSION

This secondary analysis found significant within-group increases in hemoglobin after 6 weeks of sea hare-papaya syrup administration in stunted toddlers. In contrast, serum ferritin changes were numerical and not statistically significant. These findings suggest a possible short-term hematological signal, but they do not establish that the experimental syrup is superior to commercial iron syrup or placebo.

The observed hemoglobin response may reflect improved nutrient availability or utilization during the intervention period. Erythropoiesis depends on iron as well as protein, folate, vitamin B12, and other micronutrients.³⁻⁶ Sea hare extract has been reported to contain several secondary metabolites, and papaya may contribute nutrients and bioactive compounds relevant to dietary quality.⁷⁻¹⁰ However, without batch-level nutrient composition, active-compound standardization, and precise dose reporting, the mechanism of the observed hemoglobin changes cannot be determined.

The lack of a significant ferritin response is biologically plausible. Ferritin reflects iron stores and may require a longer intervention period to show meaningful improvement. Ferritin is also an acute-phase reactant; infection or inflammation can increase ferritin independently of iron stores.^{2,5,6} Inflammatory markers were not measured in this study, which limits interpretation of ferritin as a marker of iron status.

The findings should be interpreted in relation to the previous report from the same source cohort.¹¹ That article focused on growth and motor-development outcomes, whereas the current manuscript reports hemoglobin and serum ferritin outcomes. This disclosure is important to clarify that the present analysis is related to, but distinct from, the earlier publication.

6

This study has several limitations. First, the sample size was small, with only 10 toddlers in each of five study arms, limiting statistical power and precision. Second, the available records did not include a complete sample-size calculation, detailed sequence generation, allocation-concealment documentation, or prospective trial registration. Third, no adjustment for multiple comparisons was applied. Fourth, the source data did not include inflammatory markers, complete adverse-event grading, or detailed batch-level standardization of the syrup formulations. Finally, ² the study was conducted in a limited community setting, which may restrict generalizability.

In conclusion, sea hare-papaya syrup was associated with within-group improvement in hemoglobin among stunted toddlers, while the effect on serum ferritin remained uncertain. Larger prospectively registered trials with stronger randomization procedures, standardized dosing, safety monitoring, inflammation assessment, and prespecified between-group analyses are needed before clinical effectiveness can be inferred.

20

ACKNOWLEDGEMENTS

None.

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

8

REFERENCES

1. UNICEF, World Health Organization, World Bank Group. Levels and trends in child malnutrition: UNICEF/WHO/World Bank Group joint child malnutrition estimates: key findings of the 2024 edition. New York: UNICEF, 2024.
- ¹ 2. World Health Organization. WHO guideline on use of ferritin concentrations to assess iron status in individuals and populations. Geneva: World Health Organization, 2020.

3. ¹ Petry N, Olofin I, Hurrell RF, Boy E, Wirth JP, Moursi M, et al. The proportion of anemia associated with iron deficiency in low, medium, and high human development index countries: a systematic analysis. *Nutrients* 2016;8:693.
4. ⁷ Chanpura VR, Shah NT. A study of prevalence of iron deficiency anaemia in infants and toddlers (6-24 months) by measuring serum ferritin levels. *Int J Contemp Pediatr* 2019;6:656-9.
5. ³ Bahr TM, Baer VL, Ohls RK, Christensen RD, Ward DM, Bennett ST, et al. Reconciling markedly discordant values of serum ferritin versus reticulocyte hemoglobin content. *J Perinatol* 2021;41:619-26.
6. ⁴ Fite MB, Tura AK, Yadeta TA, Oljira L, Roba KT. Prevalence, predictors of low birth weight and its association with maternal iron status using serum ferritin concentration in rural Eastern Ethiopia: a prospective cohort study. *BMC Nutr* 2022;8:70.
7. ⁹ Toruntju SA, Banudi L, Leksono P, Rahmat M, Salma WO. Identification of secondary metabolite contents on marine rabbit extract (*Dolabella auricularia*). *IOP Conf Ser Earth Environ Sci* 2020;465:012038.
8. ¹¹ Yati S, Amelia R, Octaviani DA, Erisna M. Effect of guava juice and papaya juice on increasing the hemoglobin levels among pregnant women. *Int J Nurs Health Serv* 2020;3:695-9.
9. ¹⁴ Suhartiningsih, Parida ES, Dwiatmojo NF. The difference in the effect of red guava juice and papaya juice on hemoglobin levels in adolescent girls using a hemoglobinometer. *PrimA J Ilm Ilmu Kesehat* 2025;11:116-22.
10. ¹⁷ Susanto FD, Sari LYW, Muflihah AI, Wijaya AF, Agustin AT. The effect of papaya leaf extract (*Carica papaya L.*) on erythrocyte values in anemia-model rats. *J Med Lab Infect Dis* 2025;3:1-8.
11. ¹³ Toruntju SA, Fatmawati, Gani K, Rofiqoh, Nadimin. Efficacy study of sea rabbit-papaya juice syrup on growth and motor development in stunted and malnourished toddlers. *Res J Pharm Technol* 2025;18:2602-8.

12. Handiso YH, Belachew T, Abuye C, Workicho A, Baye K. A community-based randomized controlled trial providing weekly iron-folic acid supplementation increased serum ferritin, folate and hemoglobin concentration of adolescent girls in southern Ethiopia. *Sci Rep* 2021;11:9646.

Table 1. Baseline Dietary Intake by Study Group

Group	Energy (kcal)	Protein (g)	Iron (mg)
Sea hare-papaya 80:20	1003	37.4	9.2
Sea hare-papaya 70:30	1033	39.0	10.6
Sea hare-papaya 60:40	1008	37.0	10.3
Commercial iron syrup	1040	33.2	8.9
Placebo	1119	41.3	8.6

Note: The reported overall P-value for between-group comparison of baseline dietary intake was 0.254.

Table 2. Hemoglobin Levels Before and After Intervention

Group	Before (mean±SD)	After (mean±SD)	t-value	P-value
Sea hare-papaya 80:20	10.9±0.61	11.7±0.51	-4.43	0.002
Sea hare-papaya 70:30	11.1±0.86	11.8±0.57	-3.55	0.008
Sea hare-papaya 60:40	10.8±1.23	11.6±0.45	-3.10	0.015
Commercial iron syrup	10.9±1.02	10.9±0.90	-0.73	0.486
Placebo	10.7±0.68	11.2±0.38	-2.31	0.054

Note: Values are mean±standard deviation. P-values are from paired t-tests. Hemoglobin is expressed in g/dL.

Table 3. Serum Ferritin Levels Before and After Intervention

Group	Before (mean±SD)	After (mean±SD)	t-value	P-value
Sea hare-papaya 80:20	27.54±16.82	43.45±28.47	-1.95	0.082
Sea hare-papaya 70:30	13.83±7.17	16.75±7.75	-1.78	0.109
Sea hare-papaya 60:40	14.22±15.72	19.47±13.54	-1.47	0.174
Commercial iron syrup	6.83±4.73	8.72±5.86	-0.92	0.380
Placebo	9.90±8.51	13.89±9.72	-1.34	0.213

Note: Values are mean±standard deviation. P-values are from paired t-tests. Ferritin is expressed in ng/mL.



21%

SIMILARITY INDEX

PRIMARY SOURCES

1	pmc.ncbi.nlm.nih.gov Internet	83 words — 3%
2	www.mdpi.com Internet	56 words — 2%
3	www.nature.com Internet	44 words — 1%
4	cmhnaaa.org.pe Internet	40 words — 1%
5	jurnal.poltekkespadang.ac.id Internet	40 words — 1%
6	www.academypublication.com Internet	36 words — 1%
7	www.researchgate.net Internet	32 words — 1%
8	assets-eu.researchsquare.com Internet	29 words — 1%
9	biodiversitas.mipa.uns.ac.id Internet	27 words — 1%
10	Mahmut Algül, Fatih Mete, Bostan Özlem Gayret, Meltem Erol, Gül Nihal Özdemir. "A Randomized Comparison Between Daily, Alternate Day and Weekly Single Dose Oral Iron Therapy in Children With Iron Deficiency Anemia", Journal of Pediatric Hematology/Oncology, 2025 Crossref	25 words — 1%
11	ocs.unism.ac.id Internet	23 words — 1%

12	dspace.uib.es Internet	20 words — 1%
13	rjptonline.org Internet	20 words — 1%
14	jurnal.stikes-mataram.ac.id Internet	18 words — 1%
15	www.researchsquare.com Internet	17 words — 1%
16	iris.unito.it Internet	16 words — 1%
17	jmid.uds.ac.id Internet	16 words — 1%
18	spiral.imperial.ac.uk Internet	14 words — < 1%
19	journal2.uad.ac.id Internet	12 words — < 1%
20	medcraveonline.com Internet	12 words — < 1%
21	www.medrxiv.org Internet	12 words — < 1%
22	link.springer.com Internet	10 words — < 1%
23	www.jcdr.net Internet	10 words — < 1%
24	static-curis.ku.dk Internet	9 words — < 1%

EXCLUDE QUOTES OFF
EXCLUDE BIBLIOGRAPHY OFF

EXCLUDE SOURCES OFF
EXCLUDE MATCHES OFF